

Package ‘ShrinkageTrees’

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Description Bayesian regression tree ensembles for survival analysis and causal inference. Implements BART, DART, Bayesian Causal Forests (BCF), and Horseshoe Forest models. Supports right-censored and interval-censored survival outcomes via accelerated failure time (AFT) formulations. Designed for high-dimensional prediction and heterogeneous treatment effect estimation.

URL <https://github.com/tijn-jacobs/ShrinkageTrees>

BugReports <https://github.com/tijn-jacobs/ShrinkageTrees/issues>

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Contents

ShrinkageTrees-package	2
as.mcmc.list.ShrinkageTrees	3
bayesian_bootstrap_ate	4
CausalHorseForest	5
CausalShrinkageForest	11
HorseTrees	17
ovarian	21
ovarian_truth	23
pdac	23
plot.CausalShrinkageForest	25
plot.ShrinkageTrees	26
plot.ShrinkageTreesPrediction	28
predict.CausalShrinkageForest	29
predict.ShrinkageTrees	31
print.CausalShrinkageForest	32
print.CausalShrinkageForestPrediction	32
print.ShrinkageTrees	33
print.ShrinkageTreesPrediction	33
print.summary.CausalShrinkageForest	34
print.summary.CausalShrinkageForestPrediction	35
print.summary.ShrinkageTrees	35
print.summary.ShrinkageTreesPrediction	36
ShrinkageTrees	36
summary.CausalShrinkageForest	41
summary.CausalShrinkageForestPrediction	42
summary.ShrinkageTrees	43
summary.ShrinkageTreesPrediction	44
SurvivalBART	44
SurvivalBCF	47
SurvivalDART	50
SurvivalShrinkageBCF	52

ShrinkageTrees-package

ShrinkageTrees: Bayesian Tree Ensembles for Survival Analysis and Causal Inference

Description

Bayesian regression tree ensembles for survival analysis and causal inference. Implements BART, DART, Bayesian Causal Forests (BCF), and Horseshoe Forest models. Supports right-censored and interval-censored survival outcomes via accelerated failure time (AFT) formulations. Designed for high-dimensional prediction and heterogeneous treatment effect estimation.

Author(s)

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See Also

Useful links:

- <https://github.com/tijn-jacobs/ShrinkageTrees>
- Report bugs at <https://github.com/tijn-jacobs/ShrinkageTrees/issues>

as.mcmc.list.ShrinkageTrees

Convert MCMC output to a coda mcmc.list

Description

Converts the posterior draws stored in a ShrinkageTrees object into a `mcmc.list` for use with the **coda** package's convergence diagnostics (Gelman–Rubin $\hat{R}$, effective sample size, Geweke test, etc.).

Usage

```
## S3 method for class 'ShrinkageTrees'
as.mcmc.list(x, ...)
```

Arguments

<code>x</code>	A fitted ShrinkageTrees object.
<code>...</code>	Currently unused.

Details

Requires the suggested package **coda**. For single-chain fits the returned object contains one chain.

Value

A `mcmc.list` object. Each chain is an `mcmc` object whose columns include:

sigma Posterior draws of the residual standard deviation (continuous and survival outcomes only).

See Also

`summary.ShrinkageTrees` which reports R-hat and ESS automatically when coda is available.

Examples

```
fit <- HorseTrees(y = rnorm(50), X_train = matrix(rnorm(250), 50, 5),
                N_post = 200, N_burn = 100, n_chains = 2)
if (requireNamespace("coda", quietly = TRUE)) {
  mcmc_obj <- coda::as.mcmc.list(fit)
  coda::gelman.diag(mcmc_obj)
  coda::effectiveSize(mcmc_obj)
}
```

bayesian_bootstrap_ate

Bayesian bootstrap average treatment effect

Description

Post-hoc reweights the stored posterior CATE draws of a fitted causal model to produce credible intervals for the *population* ATE (PATE) that incorporate uncertainty in the covariate distribution F_X .

Usage

```
bayesian_bootstrap_ate(object, alpha = 0.05)
```

Arguments

object	Either a fitted CausalShrinkageForest (from CausalShrinkageForest or CausalHorseForest with store_posterior_sample = TRUE) or a CausalShrinkageForestPrediction (from predict.CausalShrinkageForest).
alpha	One minus the credible level. Default 0.05 (a 95 percent credible interval).

Details

At each MCMC iteration s the conditional treatment effects $\tau^{(s)}(x_i)$ are reweighted with $(w_1^{(s)}, \dots, w_n^{(s)}) \sim \text{Dir}(1, \dots, 1)$ to give a draw

$$\widehat{\text{PATE}}^{(s)} = \sum_{i=1}^n w_i^{(s)} \tau^{(s)}(x_i).$$

The collection $\{\widehat{\text{PATE}}^{(s)}\}$ approximates the posterior of the PATE, integrating over $\tau(\cdot)$ and F_X . The equal-weight mixed ATE (MATE), $\widehat{\text{MATE}}^{(s)} = n^{-1} \sum_i \tau^{(s)}(x_i)$, is returned alongside for comparison.

For reproducibility, call `set.seed()` before invoking the function to fix the Dirichlet draws.

Value

A list with

pate_mean, pate_ci, pate_samples Posterior mean, credible interval (named lower and upper), and full vector of draws of the Bayesian-bootstrap PATE.

mate_mean, mate_ci, mate_samples Same quantities for the equal-weight mixed ATE.

n, S Number of observations and posterior draws used.

See Also

[summary.CausalShrinkageForest](#), [plot.CausalShrinkageForest](#), [predict.CausalShrinkageForest](#)

Examples

```
# Small toy causal model (binary outcome, for speed)
set.seed(1)
n <- 40; p <- 3
X <- matrix(runif(n * p), ncol = p)
trt <- rbinom(n, 1, 0.5)
y <- X[, 1] + trt * (0.5 + X[, 2]) + rnorm(n)

fit <- CausalShrinkageForest(
  y = y,
  X_train_control = X, X_train_treat = X,
  treatment_indicator_train = trt,
  outcome_type = "continuous",
  number_of_trees_control = 5, number_of_trees_treat = 5,
  prior_type_control = "horseshoe", prior_type_treat = "horseshoe",
  local_hp_control = 0.1, global_hp_control = 0.1,
  local_hp_treat = 0.1, global_hp_treat = 0.1,
  N_post = 20, N_burn = 10,
  store_posterior_sample = TRUE,
  verbose = FALSE
)

bb <- bayesian_bootstrap_ate(fit, alpha = 0.05)
bb$pate_mean
bb$pate_ci
```

CausalHorseForest

Causal Horseshoe Forests

Description

This function fits a (Bayesian) Causal Horseshoe Forest. It can be used for estimation of conditional average treatments effects of survival data given high-dimensional covariates. The outcome is decomposed in a prognostic part (control) and a treatment effect part. For both of these, we specify a Horseshoe Trees regression function. Supports continuous, right-censored, and interval-censored outcomes.

Usage

```
CausalHorseForest(
  y = NULL,
  status = NULL,
  X_train_control,
  X_train_treat,
  treatment_indicator_train,
  X_test_control = NULL,
  X_test_treat = NULL,
  treatment_indicator_test = NULL,
  left_time = NULL,
  right_time = NULL,
  outcome_type = "continuous",
  timescale = "time",
```

```

number_of_trees = 200,
k = 0.1,
power = 2,
base = 0.95,
p_grow = 0.4,
p_prune = 0.4,
nu = 3,
q = 0.9,
sigma = NULL,
N_post = 5000,
N_burn = 5000,
delayed_proposal = 5,
store_posterior_sample = FALSE,
treatment_coding = "centered",
propensity = NULL,
propensity_test = NULL,
n_chains = 1,
verbose = TRUE
)

```

Arguments

<code>y</code>	Outcome vector. For survival, represents follow-up times (can be on original or log scale depending on timescale). Set to NULL when using <code>outcome_type = "interval-censored"</code> , as values are derived from <code>left_time</code> and <code>right_time</code> .
<code>status</code>	Optional event indicator vector (1 = event occurred, 0 = censored). Required when <code>outcome_type = "right-censored"</code> . For interval-censored outcomes, this is derived automatically from <code>left_time</code> and <code>right_time</code> .
<code>X_train_control</code>	Covariate matrix for the control forest. Rows correspond to samples, columns to covariates.
<code>X_train_treat</code>	Covariate matrix for the treatment forest. Rows correspond to samples, columns to covariates.
<code>treatment_indicator_train</code>	Vector indicating treatment assignment for training samples (1 = treated, 0 = control).
<code>X_test_control</code>	Optional test covariate matrix for control forest. If NULL, defaults to column means of <code>X_train_control</code> .
<code>X_test_treat</code>	Optional test covariate matrix for treatment forest. If NULL, defaults to column means of <code>X_train_treat</code> .
<code>treatment_indicator_test</code>	Optional vector indicating treatment assignment for test samples.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> .
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Use <code>Inf</code> for right-censored observations.
<code>outcome_type</code>	Type of outcome: one of "continuous", "right-censored", or "interval-censored". Default is "continuous".

timescale	For survival outcomes: either "time" (original time scale, log-transformed internally) or "log" (already log-transformed). Used when outcome_type is "right-censored" or "interval-censored".
number_of_trees	Number of trees in each forest. Default is 200.
k	Horseshoe prior scale hyperparameter. Default is 0.1. Controls global-local shrinkage on step heights.
power	Power parameter for tree structure prior. Default is 2.0.
base	Base parameter for tree structure prior. Default is 0.95.
p_grow	Probability of proposing a grow move. Default is 0.4.
p_prune	Probability of proposing a prune move. Default is 0.4.
nu	Degrees of freedom for the error variance prior. Default is 3.
q	Quantile parameter for error variance prior. Default is 0.90.
sigma	Optional known standard deviation of the outcome. If NULL, estimated from data.
N_post	Number of posterior samples to store. Default is 5000.
N_burn	Number of burn-in iterations. Default is 5000.
delayed_proposal	Number of delayed iterations before proposal updates. Default is 5.
store_posterior_sample	Logical; whether to store posterior samples of predictions. Default is FALSE.
treatment_coding	Treatment coding scheme for the two-forest model. One of "centered" (default), "binary", "adaptive", or "invariant". "centered" uses $b_i \in \{-1/2, 1/2\}$; "binary" uses $b_i \in \{0, 1\}$; "adaptive" uses $b_i = A_i - \hat{e}(x_i)$ where $\hat{e}(x_i)$ is the estimated propensity score; "invariant" treats b_0, b_1 as parameters estimated within the Gibbs sampler with $b_j \sim N(0, 1/2)$ priors, yielding a parameterisation-invariant model (Hahn et al., 2020, Section 5.2).
propensity	Optional numeric vector of propensity scores $\hat{e}(x_i)$ for training observations. Required when treatment_coding = "adaptive".
propensity_test	Optional numeric vector of propensity scores for test observations. Only used when treatment_coding = "adaptive". Defaults to 0.5 for all test observations if not provided.
n_chains	Number of independent MCMC chains to run. Default is 1 (standard single-chain behaviour). When n_chains > 1 the chains are run in parallel via <code>parallel::mclapply</code> and their posterior samples are pooled into a single <code>CausalShrinkageForest</code> object, so all existing print and summary methods work without modification. On Windows, <code>mclapply</code> falls back to sequential execution.
verbose	Logical; whether to print verbose output during sampling. Default is TRUE.

Details

The model separately regularizes the control and treatment trees using Horseshoe priors with global-local shrinkage on the step heights. This approach is designed for robust estimation of heterogeneous treatment effects in high-dimensional settings. It supports continuous, right-censored, and interval-censored survival outcomes. For interval-censored data, provide `left_time` and `right_time` instead of `y` and `status`; the event indicators are derived internally following the `survival::Surv(type = "interval2")` convention.

Value

An S3 object of class "CausalShrinkageForest" containing:

train_predictions Posterior mean predictions on training data (combined forest).

test_predictions Posterior mean predictions on test data (combined forest).

train_predictions_control Estimated control outcomes on training data.

test_predictions_control Estimated control outcomes on test data.

train_predictions_treat Estimated treatment effects on training data.

test_predictions_treat Estimated treatment effects on test data.

sigma Vector of posterior samples for the error standard deviation.

acceptance_ratio_control Average acceptance ratio in control forest.

acceptance_ratio_treat Average acceptance ratio in treatment forest.

train_predictions_sample_control Matrix of posterior samples for control predictions (if `store_posterior_sample = TRUE`).

test_predictions_sample_control Matrix of posterior samples for control predictions (if `store_posterior_sample = TRUE`).

train_predictions_sample_treat Matrix of posterior samples for treatment effects (if `store_posterior_sample = TRUE`).

test_predictions_sample_treat Matrix of posterior samples for treatment effects (if `store_posterior_sample = TRUE`).

See Also

Model family: [HorseTrees](#) (non-causal, horseshoe prior), [ShrinkageTrees](#) (non-causal, flexible prior), [CausalShrinkageForest](#) (causal, flexible prior).

Survival wrappers: [SurvivalBCF](#), [SurvivalShrinkageBCF](#).

S3 methods: [print.CausalShrinkageForest](#), [summary.CausalShrinkageForest](#), [predict.CausalShrinkageForest](#), [plot.CausalShrinkageForest](#).

Examples

```
# Example: Continuous outcome and homogeneous treatment effect
n <- 50
p <- 3
X_control <- matrix(runif(n * p), ncol = p)
X_treat <- matrix(runif(n * p), ncol = p)
treatment <- rbinom(n, 1, 0.5)
tau <- 2
y <- X_control[, 1] + (0.5 - treatment) * tau + rnorm(n)

fit <- CausalHorseForest(
  y = y,
  X_train_control = X_control,
  X_train_treat = X_treat,
  treatment_indicator_train = treatment,
  outcome_type = "continuous",
  number_of_trees = 5,
  N_post = 10,
  N_burn = 5,
  store_posterior_sample = TRUE,
```



```

    verbose = FALSE
  )

  ## Example: Right-censored survival outcome
  # Set data dimensions
  n <- 100
  p <- 1000

  # Generate covariates
  X <- matrix(runif(n * p), ncol = p)
  X_treat <- X
  treatment <- rbinom(n, 1, pnorm(X[, 1] - 1/2))

  # Generate true survival times depending on X and treatment
  linpred <- X[, 1] - X[, 2] + (treatment - 0.5) * (1 + X[, 2] / 2 + X[, 3] / 3
                                                    + X[, 4] / 4)
  true_time <- linpred + rnorm(n, 0, 0.5)

  # Generate censoring times
  censor_time <- log(rexp(n, rate = 1 / 5))

  # Observed times and event indicator
  time_obs <- pmin(true_time, censor_time)
  status <- as.numeric(true_time == time_obs)

  # Estimate propensity score using HorseTrees
  fit_prop <- HorseTrees(
    y = treatment,
    X_train = X,
    outcome_type = "binary",
    number_of_trees = 200,
    N_post = 1000,
    N_burn = 1000
  )

  # Retrieve estimated probability of treatment (propensity score)
  propensity <- fit_prop$train_probabilities

  # Combine propensity score with covariates for control forest
  X_control <- cbind(propensity, X)

  # Fit the Causal Horseshoe Forest for survival outcome
  fit_surv <- CausalHorseForest(
    y = time_obs,
    status = status,
    X_train_control = X_control,
    X_train_treat = X_treat,
    treatment_indicator_train = treatment,
    outcome_type = "right-censored",
    timescale = "log",
    number_of_trees = 200,
    k = 0.1,
    N_post = 1000,
    N_burn = 1000,
    store_posterior_sample = TRUE
  )

```

```

## Evaluate and summarize results

# Evaluate C-index if survival package is available
if (requireNamespace("survival", quietly = TRUE)) {
  predicted_survtime <- fit_surv$train_predictions
  cindex_result <- survival::concordance(survival::Surv(time_obs, status) ~ predicted_survtime)
  c_index <- cindex_result$concordance
  cat("C-index:", round(c_index, 3), "\n")
} else {
  cat("Package 'survival' not available. Skipping C-index computation.\n")
}

# Compute posterior ATE samples
ate_samples <- rowMeans(fit_surv$train_predictions_sample_treat)
mean_ate <- mean(ate_samples)
ci_95 <- quantile(ate_samples, probs = c(0.025, 0.975))

cat("Posterior mean ATE:", round(mean_ate, 3), "\n")
cat("95% credible interval: [", round(ci_95[1], 3), ", ", round(ci_95[2], 3), "]\n", sep = "")

# Plot histogram of ATE samples
hist(
  ate_samples,
  breaks = 30,
  col = "steelblue",
  freq = FALSE,
  border = "white",
  xlab = "Average Treatment Effect (ATE)",
  main = "Posterior distribution of ATE"
)
abline(v = mean_ate, col = "orange3", lwd = 2)
abline(v = ci_95, col = "orange3", lty = 2, lwd = 2)
abline(v = 1.541667, col = "darkred", lwd = 2)
legend(
  "topright",
  legend = c("Mean", "95% CI", "Truth"),
  col = c("orange3", "orange3", "red"),
  lty = c(1, 2, 1),
  lwd = 2
)

## Plot individual CATE estimates

# Summarize posterior distribution per patient
posterior_matrix <- fit_surv$train_predictions_sample_treat
posterior_mean <- colMeans(posterior_matrix)
posterior_ci <- apply(posterior_matrix, 2, quantile, probs = c(0.025, 0.975))

df_cate <- data.frame(
  mean = posterior_mean,
  lower = posterior_ci[1, ],
  upper = posterior_ci[2, ]
)

# Sort patients by posterior mean CATE
df_cate_sorted <- df_cate[order(df_cate$mean), ]

```

```

n_patients <- nrow(df_cate_sorted)

# Create the plot
plot(
  x = df_cate_sorted$mean,
  y = 1:n_patients,
  type = "n",
  xlab = "CATE per patient (95% credible interval)",
  ylab = "Patient index (sorted)",
  main = "Posterior CATE estimates",
  xlim = range(df_cate_sorted$lower, df_cate_sorted$upper)
)

# Add CATE intervals
segments(
  x0 = df_cate_sorted$lower,
  x1 = df_cate_sorted$upper,
  y0 = 1:n_patients,
  y1 = 1:n_patients,
  col = "steelblue"
)

# Add mean points
points(df_cate_sorted$mean, 1:n_patients, pch = 16, col = "orange3", lwd = 0.1)

# Add reference line at 0
abline(v = 0, col = "black", lwd = 2)

```

CausalShrinkageForest *General Causal Shrinkage Forests*

Description

Fits a (Bayesian) Causal Shrinkage Forest model for estimating heterogeneous treatment effects. This function generalizes [CausalHorseForest](#) by allowing flexible global-local shrinkage priors on the step heights in both the control and treatment forests. It supports continuous, right-censored, and interval-censored survival outcomes.

Usage

```

CausalShrinkageForest(
  y = NULL,
  status = NULL,
  X_train_control,
  X_train_treat,
  treatment_indicator_train,
  X_test_control = NULL,
  X_test_treat = NULL,
  treatment_indicator_test = NULL,
  left_time = NULL,

```

```

right_time = NULL,
outcome_type = "continuous",
timescale = "time",
number_of_trees_control = 200,
number_of_trees_treat = 200,
prior_type_control = "horseshoe",
prior_type_treat = "horseshoe",
local_hp_control = NULL,
local_hp_treat = NULL,
global_hp_control = NULL,
global_hp_treat = NULL,
a_dirichlet_control = 0.5,
a_dirichlet_treat = 0.5,
b_dirichlet_control = 1,
b_dirichlet_treat = 1,
rho_dirichlet_control = NULL,
rho_dirichlet_treat = NULL,
power_control = 2,
power_treat = 2,
base_control = 0.95,
base_treat = 0.95,
p_grow = 0.5,
p_prune = 0.5,
nu = 3,
q = 0.9,
sigma = NULL,
N_post = 5000,
N_burn = 5000,
delayed_proposal = 5,
store_posterior_sample = FALSE,
treatment_coding = "centered",
propensity = NULL,
propensity_test = NULL,
n_chains = 1,
verbose = TRUE
)

```

Arguments

<code>y</code>	Outcome vector. Numeric. Represents continuous outcomes or follow-up times. Set to NULL when using <code>outcome_type = "interval-censored"</code> , as values are derived from <code>left_time</code> and <code>right_time</code> .
<code>status</code>	Optional event indicator vector (1 = event occurred, 0 = censored). Required when <code>outcome_type = "right-censored"</code> . For interval-censored outcomes, this is derived automatically from <code>left_time</code> and <code>right_time</code> .
<code>X_train_control</code>	Covariate matrix for the control forest. Rows correspond to samples, columns to covariates.
<code>X_train_treat</code>	Covariate matrix for the treatment forest.
<code>treatment_indicator_train</code>	Vector indicating treatment assignment for training samples (1 = treated, 0 = control).

<code>X_test_control</code>	Optional covariate matrix for control forest test data. Defaults to column means of <code>X_train_control</code> if NULL.
<code>X_test_treat</code>	Optional covariate matrix for treatment forest test data. Defaults to column means of <code>X_train_treat</code> if NULL.
<code>treatment_indicator_test</code>	Optional vector indicating treatment assignment for test data.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> .
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Use <code>Inf</code> for right-censored observations.
<code>outcome_type</code>	Type of outcome: one of "continuous", "right-censored", or "interval-censored". Default is "continuous".
<code>timescale</code>	For survival outcomes: either "time" (original scale, log-transformed internally) or "log" (already log-transformed). Default is "time". Used when <code>outcome_type</code> is "right-censored" or "interval-censored".
<code>number_of_trees_control</code>	Number of trees in the control forest. Default is 200.
<code>number_of_trees_treat</code>	Number of trees in the treatment forest. Default is 200.
<code>prior_type_control</code>	Type of prior on control forest step heights. One of "horseshoe", "horseshoe_fw", or "half-cauchy". Default is "horseshoe".
<code>prior_type_treat</code>	Type of prior on treatment forest step heights. Same options as <code>prior_type_control</code> .
<code>local_hp_control</code>	Local hyperparameter controlling shrinkage on individual steps (control forest). Required for all prior types.
<code>local_hp_treat</code>	Local hyperparameter for treatment forest.
<code>global_hp_control</code>	Global hyperparameter for control forest. Required for horseshoe-type priors; ignored for "half-cauchy".
<code>global_hp_treat</code>	Global hyperparameter for treatment forest.
<code>a_dirichlet_control</code>	First shape parameter of the Beta prior used in the Dirichlet-Sparse splitting rule for the control forest. Together with <code>b_dirichlet_control</code> , it controls the expected sparsity level.
<code>a_dirichlet_treat</code>	First shape parameter of the Beta prior used in the Dirichlet-Sparse splitting rule for the treatment forest.
<code>b_dirichlet_control</code>	Second shape parameter of the Beta prior for the sparsity level in the control forest. Larger values shrink splitting probabilities more strongly toward uniform sparsity.
<code>b_dirichlet_treat</code>	Second shape parameter of the Beta prior governing sparsity in the treatment forest.

rho_dirichlet_control	Sparsity hyperparameter for the control forest. Represents the <i>expected number of active predictors</i> . If left NULL, it defaults to the number of covariates in the control forest.
rho_dirichlet_treat	Sparsity hyperparameter for the treatment forest, interpreted as the expected number of active predictors. Defaults to the number of covariates in the treatment forest if not specified.
power_control	Power parameter for the control forest tree structure prior splitting probability.
power_treat	Power parameter for the treatment forest tree structure prior splitting probability.
base_control	Base parameter for the control forest tree structure prior splitting probability.
base_treat	Base parameter for the treatment forest tree structure prior splitting probability.
p_grow	Probability of proposing a grow move. Default is 0.5. These are fixed at 0.5 for prior_type "standard" and "dirichlet".
p_prune	Probability of proposing a prune move. Default is 0.5. These are fixed at 0.5 for prior_type "standard" and "dirichlet".
nu	Degrees of freedom for the error variance prior. Default is 3.
q	Quantile parameter for error variance prior. Default is 0.90.
sigma	Optional known standard deviation of the outcome. If NULL, estimated from data.
N_post	Number of posterior samples to store. Default is 5000.
N_burn	Number of burn-in iterations. Default is 5000.
delayed_proposal	Number of delayed iterations before proposal updates. Default is 5.
store_posterior_sample	Logical; whether to store posterior samples of predictions. Default is FALSE.
treatment_coding	Treatment coding scheme for the two-forest model. One of "centered" (default), "binary", "adaptive", or "invariant". "centered" uses $b_i \in \{-1/2, 1/2\}$; "binary" uses $b_i \in \{0, 1\}$; "adaptive" uses $b_i = A_i - \hat{e}(x_i)$ where $\hat{e}(x_i)$ is the estimated propensity score; "invariant" treats b_0, b_1 as parameters estimated within the Gibbs sampler with $b_j \sim N(0, 1/2)$ priors, yielding a parameterisation-invariant model (Hahn et al., 2020, Section 5.2).
propensity	Optional numeric vector of propensity scores $\hat{e}(x_i)$ for training observations. Required when treatment_coding = "adaptive".
propensity_test	Optional numeric vector of propensity scores for test observations. Only used when treatment_coding = "adaptive". Defaults to 0.5 for all test observations if not provided.
n_chains	Number of independent MCMC chains to run. Default is 1 (standard single-chain behaviour). When n_chains > 1 the chains are run in parallel via <code>parallel::mclapply</code> and their posterior samples are pooled into a single CausalShrinkageForest object, so all existing print and summary methods work without modification. On Windows, mclapply falls back to sequential execution.
verbose	Logical; whether to print verbose output. Default is TRUE.

Details

This function is a flexible generalization of CausalHorseForest. The Causal Shrinkage Forest model decomposes the outcome into a prognostic (control) and a treatment effect part. Each part is modeled by its own shrinkage tree ensemble, with separate flexible global-local shrinkage priors. It is particularly useful for estimating heterogeneous treatment effects in high-dimensional settings. Further methodological details on the Horseshoe Forest framework can be found in Jacobs, van Wieringen & van der Pas (2025).

The horseshoe prior is the fully Bayesian global-local shrinkage prior, where both the global and local shrinkage parameters are assigned half-Cauchy distributions with scale hyperparameters `global_hp` and `local_hp`, respectively. The global shrinkage parameter is defined separately for each tree, allowing adaptive regularization per tree.

The horseshoe_fw prior (forest-wide horseshoe) is similar to horseshoe, except that the global shrinkage parameter is shared across all trees in the forest simultaneously.

The half-cauchy prior considers only local shrinkage and does not include a global shrinkage component. It places a half-Cauchy prior on each local shrinkage parameter with scale hyperparameter `local_hp`.

The dirichlet prior implements the Dirichlet-Sparse splitting rule of Linero (2018), in which splitting probabilities follow a Dirichlet prior whose concentration is controlled by a Beta sparsity parameter (`a_dirichlet`, `b_dirichlet`) and an expected sparsity level `rho_dirichlet`.

Value

An S3 object of class "CausalShrinkageForest" containing:

train_predictions Posterior mean predictions on training data (combined forest).

test_predictions Posterior mean predictions on test data (combined forest).

train_predictions_control Estimated control outcomes on training data.

test_predictions_control Estimated control outcomes on test data.

train_predictions_treat Estimated treatment effects on training data.

test_predictions_treat Estimated treatment effects on test data.

sigma Vector of posterior samples for the error standard deviation.

acceptance_ratio_control Average acceptance ratio in control forest.

acceptance_ratio_treat Average acceptance ratio in treatment forest.

train_predictions_sample_control Matrix of posterior samples for control predictions (if `store_posterior_sample = TRUE`).

test_predictions_sample_control Matrix of posterior samples for control predictions (if `store_posterior_sample = TRUE`).

train_predictions_sample_treat Matrix of posterior samples for treatment effects (if `store_posterior_sample = TRUE`).

test_predictions_sample_treat Matrix of posterior samples for treatment effects (if `store_posterior_sample = TRUE`).

References

- Jacobs, T., van Wieringen, W. N., & van der Pas, S. L. (2025). *Horseshoe Forests for High-Dimensional Causal Survival Analysis*. arXiv:2507.22004. <https://doi.org/10.48550/arXiv.2507.22004>
- Chipman, H. A., George, E. I., & McCulloch, R. E. (2010). *BART: Bayesian additive regression trees*. Annals of Applied Statistics.

Linero, A. R. (2018). *Bayesian regression trees for high-dimensional prediction and variable selection*. Journal of the American Statistical Association.

See Also

Model family: [CausalHorseForest](#) (causal, horseshoe prior), [ShrinkageTrees](#) (non-causal, flexible prior), [HorseTrees](#) (non-causal, horseshoe prior).

Survival wrappers: [SurvivalBCF](#), [SurvivalShrinkageBCF](#).

S3 methods: [print.CausalShrinkageForest](#), [summary.CausalShrinkageForest](#), [predict.CausalShrinkageForest](#), [plot.CausalShrinkageForest](#).

Examples

```
# Example: Continuous outcome, homogeneous treatment effect, two priors
n <- 50
p <- 3
X <- matrix(runif(n * p), ncol = p)
X_treat <- X_control <- X
treat <- rbinom(n, 1, X[,1])
tau <- 2
y <- X[, 1] + (0.5 - treat) * tau + rnorm(n)

# Fit a standard Causal Horseshoe Forest
fit_horseshoe <- CausalShrinkageForest(y = y,
  X_train_control = X_control,
  X_train_treat = X_treat,
  treatment_indicator_train = treat,
  outcome_type = "continuous",
  number_of_trees_treat = 5,
  number_of_trees_control = 5,
  prior_type_control = "horseshoe",
  prior_type_treat = "horseshoe",
  local_hp_control = 0.1/sqrt(5),
  local_hp_treat = 0.1/sqrt(5),
  global_hp_control = 0.1/sqrt(5),
  global_hp_treat = 0.1/sqrt(5),
  N_post = 10,
  N_burn = 5,
  store_posterior_sample = TRUE,
  verbose = FALSE
)

# Fit a Causal Shrinkage Forest with half-cauchy prior
fit_halfcauchy <- CausalShrinkageForest(y = y,
  X_train_control = X_control,
  X_train_treat = X_treat,
  treatment_indicator_train = treat,
  outcome_type = "continuous",
  number_of_trees_treat = 5,
  number_of_trees_control = 5,
  prior_type_control = "half-cauchy",
  prior_type_treat = "half-cauchy",
  local_hp_control = 1/sqrt(5),
  local_hp_treat = 1/sqrt(5),
  N_post = 10,
  N_burn = 5,
```



```

                                store_posterior_sample = TRUE,
                                verbose = FALSE
)

# Posterior mean CATEs
CATE_horseshoe <- colMeans(fit_horseshoe$train_predictions_sample_treat)
CATE_halfcauchy <- colMeans(fit_halfcauchy$train_predictions_sample_treat)

# Posteriors of the ATE
post_ATE_horseshoe <- rowMeans(fit_horseshoe$train_predictions_sample_treat)
post_ATE_halfcauchy <- rowMeans(fit_halfcauchy$train_predictions_sample_treat)

# Posterior mean ATE
ATE_horseshoe <- mean(post_ATE_horseshoe)
ATE_halfcauchy <- mean(post_ATE_halfcauchy)

# Example: Interval-censored causal survival outcome
n <- 50; p <- 3
X_ic <- matrix(rnorm(n * p), ncol = p)
treat_ic <- rbinom(n, 1, 0.5)
true_t <- rexp(n, rate = exp(-X_ic[, 1] - 0.5 * treat_ic))
left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
exact <- sample(n, 15)
left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 10); right_t[rc] <- Inf

fit_ic <- CausalShrinkageForest(
  left_time = left_t, right_time = right_t,
  X_train_control = X_ic, X_train_treat = X_ic,
  treatment_indicator_train = treat_ic,
  outcome_type = "interval-censored",
  number_of_trees_control = 5, number_of_trees_treat = 5,
  prior_type_control = "horseshoe", prior_type_treat = "horseshoe",
  local_hp_control = 0.1/sqrt(5), local_hp_treat = 0.1/sqrt(5),
  global_hp_control = 0.1/sqrt(5), global_hp_treat = 0.1/sqrt(5),
  N_post = 10, N_burn = 5,
  store_posterior_sample = TRUE, verbose = FALSE)

```

HorseTrees

Horseshoe Regression Trees (HorseTrees)

Description

Fits a Bayesian Horseshoe Trees model with a single learner. Implements regularization on the step heights using a global-local Horseshoe prior, controlled via the parameter k . Supports continuous, binary, right-censored, and interval-censored (survival) outcomes.

Usage

```

HorseTrees(
  y = NULL,
  status = NULL,

```

```

X_train,
X_test = NULL,
left_time = NULL,
right_time = NULL,
outcome_type = "continuous",
timescale = "time",
number_of_trees = 200,
k = 0.1,
power = 2,
base = 0.95,
p_grow = 0.4,
p_prune = 0.4,
nu = 3,
q = 0.9,
sigma = NULL,
N_post = 1000,
N_burn = 1000,
delayed_proposal = 5,
store_posterior_sample = TRUE,
n_chains = 1,
verbose = TRUE
)

```

Arguments

<code>y</code>	Outcome vector. Numeric. Can represent continuous outcomes, binary outcomes (0/1), or follow-up times for survival data. Set to NULL (default) when using <code>outcome_type = "interval-censored"</code> , as values are derived from <code>left_time</code> and <code>right_time</code> .
<code>status</code>	Optional censoring indicator vector (1 = event occurred, 0 = censored). Required if <code>outcome_type = "right-censored"</code> . For interval-censored outcomes, this is derived automatically from <code>left_time</code> and <code>right_time</code> .
<code>X_train</code>	Covariate matrix for training. Each row corresponds to an observation, and each column to a covariate.
<code>X_test</code>	Optional covariate matrix for test data. If NULL, defaults to the mean of the training covariates.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> .
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Use <code>Inf</code> for right-censored observations.
<code>outcome_type</code>	Type of outcome. One of "continuous", "binary", "right-censored", or "interval-censored".
<code>timescale</code>	Indicates the scale of follow-up times. Options are "time" (nonnegative follow-up times, will be log-transformed internally) or "log" (already log-transformed). Only used when <code>outcome_type = "right-censored"</code> or "interval-censored".
<code>number_of_trees</code>	Number of trees in the ensemble. Default is 200.

k	Horseshoe scale hyperparameter (default 0.1). This parameter controls the overall level of shrinkage by setting the scale for both global and local shrinkage components. The local and global hyperparameters are parameterized as $\alpha = \frac{k}{\sqrt{\text{number_of_trees}}}$ to ensure adaptive regularization across trees.
power	Power parameter for tree structure prior. Default is 2.0.
base	Base parameter for tree structure prior. Default is 0.95.
p_grow	Probability of proposing a grow move. Default is 0.4.
p_prune	Probability of proposing a prune move. Default is 0.4.
nu	Degrees of freedom for the error distribution prior. Default is 3.
q	Quantile hyperparameter for the error variance prior. Default is 0.90.
sigma	Optional known value for error standard deviation. If NULL, estimated from data.
N_post	Number of posterior samples to store. Default is 1000.
N_burn	Number of burn-in iterations. Default is 1000.
delayed_proposal	Number of delayed iterations before proposal. Only for reversible updates. Default is 5.
store_posterior_sample	Logical; whether to store posterior samples for each iteration. Default is TRUE.
n_chains	Number of independent MCMC chains to run. Default is 1 (standard single-chain behaviour). When <code>n_chains > 1</code> the chains are run in parallel via <code>parallel::mclapply</code> and their posterior samples are pooled into a single <code>ShrinkageTrees</code> object, so all existing <code>print</code> , <code>summary</code> , and <code>predict</code> methods work without modification. On Windows, <code>mclapply</code> falls back to sequential execution.
verbose	Logical; whether to print verbose output. Default is TRUE.

Details

For continuous outcomes, the model centers and optionally standardizes the outcome using a prior guess of the standard deviation. For binary outcomes, the function uses a probit link formulation. For right-censored outcomes (survival data), the function can handle follow-up times either on the original time scale or log-transformed. For interval-censored outcomes, provide `left_time` and `right_time` instead of `y` and `status`; the event indicators are derived internally following the `survival::Surv(type = "interval2")` convention. Generalized implementation with multiple prior possibilities is given by [ShrinkageTrees](#).

Value

An S3 object of class "ShrinkageTrees" with the following elements:

train_predictions Vector of posterior mean predictions on the training data.

test_predictions Vector of posterior mean predictions on the test data (or on mean covariate vector if `X_test` not provided).

sigma Vector of posterior samples of the error variance.

acceptance_ratio Average acceptance ratio across trees during sampling.

train_predictions_sample Matrix of posterior samples of training predictions (iterations in rows, observations in columns). Present only if `store_posterior_sample = TRUE`.

test_predictions_sample Matrix of posterior samples of test predictions. Present only if `store_posterior_sample = TRUE`.

train_probabilities Vector of posterior mean probabilities on the training data (only for outcome_type = "binary").

test_probabilities Vector of posterior mean probabilities on the test data (only for outcome_type = "binary").

train_probabilities_sample Matrix of posterior samples of training probabilities (only for outcome_type = "binary" and if store_posterior_sample = TRUE).

test_probabilities_sample Matrix of posterior samples of test probabilities (only for outcome_type = "binary" and if store_posterior_sample = TRUE).

See Also

Model family: [ShrinkageTrees](#) (flexible prior choice), [CausalHorseForest](#) (causal inference), [CausalShrinkageForest](#) (causal, flexible prior).

Survival wrappers: [SurvivalBART](#), [SurvivalDART](#).

S3 methods: [print.ShrinkageTrees](#), [summary.ShrinkageTrees](#), [predict.ShrinkageTrees](#), [plot.ShrinkageTrees](#).

Examples

```
# Minimal example: continuous outcome
n <- 25
p <- 5
X <- matrix(rnorm(n * p), ncol = p)
y <- X[, 1] + rnorm(n)
fit1 <- HorseTrees(y = y, X_train = X, outcome_type = "continuous",
                  number_of_trees = 5, N_post = 75, N_burn = 25,
                  verbose = FALSE)

# Minimal example: binary outcome
X <- matrix(rnorm(n * p), ncol = p)
y <- ifelse(X[, 1] + rnorm(n) > 0, 1, 0)
fit2 <- HorseTrees(y = y, X_train = X, outcome_type = "binary",
                  number_of_trees = 5, N_post = 75, N_burn = 25,
                  verbose = FALSE)

# Minimal example: right-censored outcome
X <- matrix(rnorm(n * p), ncol = p)
time <- rexp(n, rate = 0.1)
status <- rbinom(n, 1, 0.7)
fit3 <- HorseTrees(y = time, status = status, X_train = X,
                  outcome_type = "right-censored", number_of_trees = 5,
                  N_post = 75, N_burn = 25, verbose = FALSE)

# Minimal example: interval-censored outcome
X <- matrix(rnorm(n * p), ncol = p)
true_t <- rexp(n, rate = 0.1)
left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
# Mark some as exact, some as right-censored
exact <- sample(n, 8); left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 5); right_t[rc] <- Inf
fit4 <- HorseTrees(left_time = left_t, right_time = right_t, X_train = X,
                  outcome_type = "interval-censored", number_of_trees = 5,
                  N_post = 75, N_burn = 25, verbose = FALSE)
```

```
# Larger continuous example (not run automatically)

n <- 100
p <- 100
X <- matrix(rnorm(100 * p), ncol = p)
X_test <- matrix(rnorm(50 * p), ncol = p)
y <- X[, 1] + X[, 2] - X[, 3] + rnorm(100, sd = 0.5)

fit5 <- HorseTrees(y = y,
                   X_train = X,
                   X_test = X_test,
                   outcome_type = "continuous",
                   number_of_trees = 200,
                   N_post = 2500,
                   N_burn = 2500,
                   store_posterior_sample = TRUE,
                   verbose = TRUE)

plot(fit4$sigma, type = "l", ylab = expression(sigma),
     xlab = "Iteration", main = "Sigma traceplot")

hist(fit4$train_predictions_sample[, 1],
     main = "Posterior distribution of prediction outcome individual 1",
     xlab = "Prediction", breaks = 20)
```

ovarian

Semi-synthesised TCGA Ovarian Cancer Dataset

Description

Gene expression and clinical covariates for ovarian cancer patients from The Cancer Genome Atlas (TCGA-OV), combined with semi-synthetic survival outcomes and treatment assignment. Real covariates (age, FIGO stage, tumor grade, gene expression) are retained; survival times, event indicator, and treatment assignment are simulated from a known data-generating process so that the true treatment effect is available for validation (see [ovarian_truth](#)).

Usage

```
ovarian
```

Format

A data frame with 357 rows (patients) and 1007 columns:

- **OS_time**: Numeric. Observed survival time in days (simulated).
- **OS_event**: Integer. Event indicator (simulated). 1 = event observed, 0 = right-censored.
- **treatment**: Integer. Simulated treatment assignment. 1 = carboplatin, 0 = cisplatin. Driven primarily by year_of_diagnosis as an instrumental variable (cisplatin era pre-~2000, carboplatin after).
- **age**: Integer. Age at initial pathologic diagnosis in years.

- **figo_stage**: Integer. FIGO stage coded as 2 = Stage II, 3 = Stage III, 4 = Stage IV.
- **tumor_grade**: Integer. Histologic tumor grade coded as 2 = G2, 3 = G3, 4 = G4. Rows with GX (unknown grade) were excluded.
- **year_of_diagnosis**: Integer. Year of initial pathologic diagnosis (approx. 1992–2013). Used as an instrumental variable for treatment assignment in the DGP.
- **right_time, left_time**: Numeric. Interval-censoring bounds derived from the simulated survival times, suitable for passing to the package's interval-censored survival interface (`right_time` = Inf for right-censored observations, `left_time` == `right_time` for exact events).
- **year_of_diagnosis.1**: Integer. Duplicate of `year_of_diagnosis` left in place from the data-assembly join; retained for reproducibility and may be ignored.
- **ENSG...**: Numeric. Log2(TPM + 1) normalised gene expression levels for 997 Ensembl genes (columns named by versioned Ensembl gene IDs, e.g. ENSG00000270372.1). Genes were selected as the most variable transcripts across TCGA-OV samples, ranked by median absolute deviation (MAD).

Details

RNA-seq data were downloaded from the GDC portal using the TCGAbiolinks package (STAR - Counts workflow). Expression values were normalised to TPM and log2-transformed as $\log_2(\text{TPM} + 1)$. Genes with median TPM ≤ 1 across all samples were removed prior to MAD filtering. Clinical data were obtained from the BCR Biotab clinical supplement. Treatment assignment was derived from the drug table (`clinical_drug_ov`), restricted to adjuvant (first-line) treatment records. Samples were matched between expression and clinical data using the 12-character TCGA patient barcode.

Source

<https://portal.gdc.cancer.gov/projects/TCGA-OV>

References

- Cancer Genome Atlas Research Network (2011). Integrated genomic analyses of ovarian carcinoma. *Nature*, 474, 609–615. doi:10.1038/nature10166
- Colaprico, A. et al. (2016). TCGAbiolinks: an R/Bioconductor package for integrative analysis with GDC data. *Nucleic Acids Research*, 44(8). doi:10.1093/nar/gkv1507

Examples

```
data(ovarian)

# Dimensions: patients x (6 clinical + 2000 gene columns)
dim(ovarian)

# Survival outcome
head(ovarian[, c("OS_time", "OS_event", "treatment")])

# KM plot by treatment
if (requireNamespace("survival", quietly = TRUE)) {
  library(survival)
  fit <- survfit(Surv(OS_time, OS_event) ~ treatment, data = ovarian)
  plot(fit, col = c("blue", "red"), xlab = "Time (days)", ylab = "Survival")
  legend("topright", c("Carboplatin", "Cisplatin"), col = c("blue", "red"), lty = 1)
}
```

ovarian_truth	<i>Ground-truth quantities for the semi-synthesised ovarian dataset</i>
---------------	---

Description

The simulated quantities that correspond to the [ovarian](#) dataset. Because the ovarian outcomes and treatment assignment are generated from a known data-generating process, the underlying potential outcomes, prognostic function, conditional treatment effect, and propensity score are available for validating estimators of treatment effects under right- and interval-censored survival.

Usage

```
ovarian_truth
```

Format

A data frame with one row per patient in [ovarian](#) and the following columns:

true_log_T Numeric. True (uncensored) survival time on the log scale.

true_T Numeric. True (uncensored) survival time on the original scale.

true_mu Numeric. True prognostic function $\mu(x)$ (expected log survival time at the reference treatment).

true_tau Numeric. True conditional average treatment effect $\tau(x)$ on the log-survival scale.

true_propensity Numeric. True propensity for the treated group (carboplatin) used to simulate the observed assignment.

See Also

[ovarian](#) for the observed semi-synthesised data.

Examples

```
data(ovarian)
data(ovarian_truth)
stopifnot(nrow(ovarian) == nrow(ovarian_truth))
# True (population) average treatment effect on the log-survival scale:
mean(ovarian_truth$true_tau)
```

pdac	<i>Processed TCGA PAAD dataset (pdac)</i>
------	---

Description

A reduced and cleaned subset of the TCGA pancreatic ductal adenocarcinoma (PAAD) dataset, derived from The Cancer Genome Atlas (TCGA) PAAD cohort. This version, `pdac`, is smaller and simplified for practical analyses and package examples.

Usage

```
pdac
```

Format

A data frame with rows corresponding to patients and columns as described above.

Details

This dataset was originally compiled and curated in the open-source pdacR package by Torre-Healy et al. (2023), which harmonized and integrated the TCGA PAAD gene expression and clinical data. The current version further reduces and simplifies the data for efficient modeling demonstrations and survival analyses.

The data frame includes:

- **time**: Overall survival time in months.
- **status**: Event indicator; 1 = event occurred, 0 = censored.
- **treatment**: Binary treatment indicator; 1 = radiation therapy, 0 = control.
- **age**: Age at initial pathologic diagnosis (numeric).
- **sex**: Binary sex indicator; 1 = male, 0 = female.
- **grade**: Tumor differentiation grade (ordinal; 1 = well, 2 = moderate, 3 = poor, 4 = undifferentiated).
- **tumor.cellularity**: Tumor cellularity estimate (numeric).
- **tumor.purity**: Tumor purity class (binary; 1 = high, 0 = low).
- **absolute.purity**: Absolute purity estimate (numeric).
- **moffitt.cluster**: Moffitt transcriptional subtype (binary; 1 = basal-like, 0 = classical).
- **meth.leukocyte.percent**: DNA methylation leukocyte estimate (numeric).
- **meth.purity.mode**: DNA methylation purity mode (numeric).
- **stage**: Nodal stage indicator (binary; 1 = n1, 0 = n0).
- **lymph.nodes**: Number of lymph nodes examined (numeric).
- **Driver gene columns**: Expression values of key driver genes (e.g., KRAS, TP53, CDKN2A, SMAD4, BRCA1, BRCA2).
- **Other gene columns**: Expression values of ~3,000 most variable non-driver genes (based on median absolute deviation).

Source

[doi:10.1016/j.ccell.2017.07.007](https://doi.org/10.1016/j.ccell.2017.07.007)

References

- Raphael BJ, et al. "Integrated genomic characterization of pancreatic ductal adenocarcinoma." *Cancer Cell*. 2017 Aug 14;32(2):185–203.e13. PMID: 28810144.
- Torre-Healy LA, Kawalerski RR, Oh K, et al. "Open-source curation of a pancreatic ductal adenocarcinoma gene expression analysis platform (pdacR) supports a two-subtype model." *Communications Biology*. 2023; <https://doi.org/10.1038/s42003-023-04461-6>.
- The Cancer Genome Atlas (TCGA), PAAD project, DbGaP: phs000178.

plot.CausalShrinkageForest

Plot diagnostics for a CausalShrinkageForest model

Description

Visualises posterior draws using **ggplot2**. Requires the suggested package **ggplot2**.

Usage

```
## S3 method for class 'CausalShrinkageForest'
plot(
  x,
  type = c("trace", "density", "ate", "cate", "vi"),
  forest = c("both", "control", "treat"),
  n_vi = 10,
  bayesian_bootstrap = TRUE,
  ...
)
```

Arguments

x	A CausalShrinkageForest object.
type	Character; one of: "trace" Sigma traceplot (chain mixing). "density" Overlaid posterior density of sigma per chain. "ate" Posterior density of the average treatment effect (ATE) with 95 percent credible region. Requires store_posterior_sample = TRUE. "cate" Point estimates and 95 percent credible intervals for the CATE of each training observation, sorted by posterior mean. Requires store_posterior_sample = TRUE. "vi" Posterior credible intervals for variable inclusion probabilities (Dirichlet prior only). Controlled by forest.
forest	For type = "vi": which forest to display. One of "both" (default), "control", or "treat". When "both", a named list of two ggplot2 objects is returned.
n_vi	Integer; number of top variables for type = "vi". Default 10.
bayesian_bootstrap	Logical; only used when type = "ate". If TRUE (default), the ATE posterior is computed by reweighting each iteration's CATE vector with Dirichlet(1, ..., 1) weights, giving the population ATE (PATE). If FALSE, equal 1/n weights are used, giving the mixed ATE (MATE).
...	Additional arguments (currently unused).

Value

A **ggplot2** object, or (for type = "vi" with forest = "both") a named list with elements control and treat.

Examples

```

if (requireNamespace("ggplot2", quietly = TRUE)) {
  set.seed(1)
  n <- 60; p <- 5
  X <- matrix(rnorm(n * p), ncol = p)
  w <- rbinom(n, 1, 0.5)
  y <- X[, 1] + w * 1.5 * (X[, 2] > 0) + rnorm(n, sd = 0.5)

  fit <- CausalShrinkageForest(
    y = y,
    X_train_control = X, X_train_treat = X,
    treatment_indicator_train = w,
    prior_type_control = "horseshoe", prior_type_treat = "horseshoe",
    local_hp_control = 0.1, global_hp_control = 0.1,
    local_hp_treat = 0.1, global_hp_treat = 0.1,
    number_of_trees_control = 5, number_of_trees_treat = 5,
    N_post = 50, N_burn = 25,
    store_posterior_sample = TRUE,
    verbose = FALSE
  )

  plot(fit, type = "trace")
  plot(fit, type = "ate")
  plot(fit, type = "cate")
}

```

plot.ShrinkageTrees *Plot diagnostics for a ShrinkageTrees model*

Description

Visualises posterior draws using **ggplot2**. Requires the suggested package **ggplot2**.

Usage

```

## S3 method for class 'ShrinkageTrees'
plot(
  x,
  type = c("trace", "density", "vi", "survival"),
  n_vi = 10,
  obs = NULL,
  t_grid = NULL,
  level = 0.95,
  km = FALSE,
  ...
)

```

Arguments

x A ShrinkageTrees object.

type Character; one of:

	"trace" Sigma traceplot across MCMC iterations (one line per chain). Useful for assessing chain mixing.
	"density" Overlaid posterior density of sigma, one curve per chain.
	"vi" Posterior credible intervals for variable inclusion probabilities (top n_vi predictors). Only available for Dirichlet prior models.
	"survival" Posterior survival curves $S(t x_i) = 1 - \Phi((\log t - \mu_i)/\sigma)$ with pointwise credible bands, derived from the AFT log-normal model. Only available for survival outcome types ("right-censored" or "interval-censored"). Population-averaged curve (default, obs = NULL): computes $\bar{S}(t) = n^{-1} \sum_i S(t x_i)$ at each MCMC iteration. The credible band reflects posterior uncertainty in both μ_i and σ when store_posterior_sample = TRUE, or sigma-only uncertainty otherwise. Individual curves (obs = c(1, 5, ...)): one curve per selected training observation with its own credible band. Set km = TRUE to overlay the Kaplan–Meier estimate as a non-parametric reference (population-averaged plot only).
n_vi	Integer; number of top variables to display when type = "vi". Default 10.
obs	Integer vector of training-set observation indices for individual survival curves, or NULL (default) for the population-averaged curve. Indices must be between 1 and the number of training observations. Used only when type = "survival".
t_grid	Optional numeric vector of time points (on the original time scale) at which to evaluate the survival function. If NULL (default), a grid of 200 equally spaced points is generated from the range of observed training times. Used only when type = "survival".
level	Width of the pointwise credible band for type = "survival". Default 0.95 (a 95 percent credible interval at each time point).
km	Logical; if TRUE and type = "survival" with obs = NULL, overlay the Kaplan–Meier curve (dashed black step function) as a non-parametric reference. Default FALSE. Requires the survival package. Ignored with a message when obs is not NULL.
...	Additional arguments (currently unused).

Value

A **ggplot2** object.

Examples

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  set.seed(1)
  n <- 50; p <- 5
  X <- matrix(rnorm(n * p), ncol = p)
  y <- X[, 1] + rnorm(n)

  # Fit a small continuous model
  fit <- ShrinkageTrees(
    y = y, X_train = X,
    prior_type = "horseshoe",
    local_hp = 0.1, global_hp = 0.1,
    number_of_trees = 5,
    N_post = 50, N_burn = 25,
    verbose = FALSE
  )
}
```

```
# Sigma traceplot -- check chain mixing
plot(fit, type = "trace")

# Overlaid posterior densities of sigma per chain
plot(fit, type = "density")
}
```

```
plot.ShrinkageTreesPrediction
```

Plot posterior predictive survival curves

Description

Plots posterior predictive survival curves for new observations from a `ShrinkageTreesPrediction` object. Only available for survival outcome types ("right-censored" or "interval-censored").

Usage

```
## S3 method for class 'ShrinkageTreesPrediction'
plot(x, type = "survival", obs = NULL, t_grid = NULL, level = 0.95, ...)
```

Arguments

<code>x</code>	A <code>ShrinkageTreesPrediction</code> object returned by predict.ShrinkageTrees for a survival model.
<code>type</code>	Character; currently only "survival" is supported.
<code>obs</code>	Integer vector of predicted-observation indices for individual survival curves, or NULL (default) for the population-averaged curve across all predicted observations.
<code>t_grid</code>	Optional numeric vector of time points (on the original time scale) at which to evaluate the survival function. If NULL (default), a grid of 200 equally spaced points is generated automatically.
<code>level</code>	Width of the pointwise credible band. Default 0.95.
<code>...</code>	Additional arguments (currently unused).

Value

A `ggplot2` object.

See Also

[predict.ShrinkageTrees](#), [plot.ShrinkageTrees](#)

Examples

```

if (requireNamespace("ggplot2", quietly = TRUE)) {
  set.seed(1)
  n <- 40; p <- 3
  X <- matrix(rnorm(n * p), ncol = p)
  X_test <- matrix(rnorm(10 * p), ncol = p)
  time <- rexp(n, rate = exp(0.5 * X[, 1]))
  status <- rbinom(n, 1, 0.7)

  fit_surv <- SurvivalBART(
    time = time, status = status, X_train = X,
    number_of_trees = 5, N_post = 50, N_burn = 25,
    store_posterior_sample = TRUE, verbose = FALSE
  )

  pred <- predict(fit_surv, newdata = X_test)
  plot(pred, type = "survival")
}

```

predict.CausalShrinkageForest

Posterior predictive inference for a CausalShrinkageForest model

Description

Re-runs the MCMC sampler on new covariate data using the stored training data and hyperparameters, returning posterior mean predictions and credible interval bounds for three quantities: the **prognostic function** (control-forest prediction $\mu(X)$), the **Conditional Average Treatment Effect** (CATE, $\tau(X)$), and the **total predicted outcome** ($\mu(X) + \tau(X)$).

Usage

```

## S3 method for class 'CausalShrinkageForest'
predict(
  object,
  newdata_control,
  newdata_treat,
  level = 0.95,
  bayesian_bootstrap = TRUE,
  ...
)

```

Arguments

object	A fitted CausalShrinkageForest model object.
newdata_control	A matrix of new covariates for the control forest, with the same number of columns as X_train_control at fit time.
newdata_treat	A matrix of new covariates for the treatment forest, with the same number of columns as X_train_treat at fit time. Must have the same number of rows as newdata_control.

<code>level</code>	Credible interval width. Default 0.95.
<code>bayesian_bootstrap</code>	Logical; if TRUE (default), the ATE over newdata is computed by reweighting each iteration's CATE vector with Dirichlet(1, ..., 1) weights (population ATE, PATE). If FALSE, equal 1/n weights are used (mixed ATE, MATE).
<code>...</code>	Currently unused.

Details

The causal forest decomposes the expected outcome as

$$E[Y | X] = \mu(X) + \tau(X) \cdot W,$$

where $\mu(X)$ is the prognostic function (control forest), $\tau(X)$ is the CATE (treatment forest), and W is the treatment indicator.

For **continuous** outcomes and survival with `timescale = "log"`, all three components are on the response scale: prognostic and total include the intercept shift ($+\bar{y}$), while cate is the pure additive treatment effect with no intercept.

For **survival with** `timescale = "time"`, predictions are back-transformed to the original time scale:

- prognostic: posterior expected baseline survival time $E[\exp(\mu(X))]$.
- cate: multiplicative effect on survival time $\exp(\tau(X))$; a value greater than 1 means treatment prolongs survival.
- total: posterior expected survival time under the observed treatment $E[\exp(\mu(X) + \tau(X))]$.

Value

A `CausalShrinkageForestPrediction` object with elements:

prognostic List with mean, lower, upper: posterior summaries of the prognostic function $\mu(X_{\text{new}})$.

cate List with mean, lower, upper: posterior summaries of the CATE $\tau(X_{\text{new}})$.

total List with mean, lower, upper: posterior summaries of the total outcome $\mu(X_{\text{new}}) + \tau(X_{\text{new}})$.

ate List with mean, lower, upper: posterior summary of the average treatment effect over newdata. For survival with `timescale = "time"`, reported as a multiplicative time ratio on the original scale.

cate_samples $S \times n_{\text{new}}$ matrix of posterior CATE draws on the scale reported in cate.

bayesian_bootstrap Flag indicating whether the reported ATE CI used Dirichlet reweighting.

n Number of test observations.

level Credible level used.

outcome_type Outcome type inherited from the fitted model.

timescale Timescale inherited from the fitted model.

See Also

[CausalHorseForest](#), [CausalShrinkageForest](#), [print.CausalShrinkageForestPrediction](#), [summary.CausalShrinkageForestPrediction](#)

predict.ShrinkageTrees

Posterior predictive inference for a ShrinkageTrees model

Description

Re-runs the MCMC sampler on new covariate data using the stored training data and hyperparameters, returning posterior mean predictions and credible interval bounds.

Usage

```
## S3 method for class 'ShrinkageTrees'
predict(object, newdata, level = 0.95, ...)
```

Arguments

<code>object</code>	A fitted ShrinkageTrees model object.
<code>newdata</code>	A matrix (or object coercible to one) of new covariates with the same number of columns as the training data.
<code>level</code>	Credible interval width. Default 0.95.
<code>...</code>	Currently unused.

Value

A ShrinkageTreesPrediction object with elements:

mean Posterior mean predictions (length nrow(newdata)).

lower Lower credible interval bound.

upper Upper credible interval bound.

n Number of test observations.

level Credible level used.

outcome_type Outcome type inherited from the fitted model.

timescale Timescale inherited from the fitted model (survival only).

predictions_sample (Survival only) N_post x n matrix of posterior predictive draws on the original scale.

sigma (Survival only) Posterior draws of sigma on the log-time scale (length N_post).

See Also

[HorseTrees](#), [ShrinkageTrees](#), [print.ShrinkageTreesPrediction](#), [summary.ShrinkageTreesPrediction](#), [plot.ShrinkageTreesPrediction](#)

```
print.CausalShrinkageForest
```

Print a CausalShrinkageForest model

Description

Displays a concise summary of a fitted CausalShrinkageForest model with per-forest columns for priors, tree counts, feature counts, and MCMC acceptance ratios.

Usage

```
## S3 method for class 'CausalShrinkageForest'
print(x, ...)
```

Arguments

x	A fitted CausalShrinkageForest model object.
...	Currently unused.

Value

Invisibly returns x.

See Also

[summary.CausalShrinkageForest](#), [CausalHorseForest](#), [CausalShrinkageForest](#)

```
print.CausalShrinkageForestPrediction
```

Print a CausalShrinkageForestPrediction object

Description

Displays a formatted table of posterior mean predictions and credible interval bounds for the first n_head observations, with separate sections for the prognostic function $\mu(X)$, the CATE $\tau(X)$, and the total outcome $\mu(X) + \tau(X)$.

Usage

```
## S3 method for class 'CausalShrinkageForestPrediction'
print(x, n_head = 6, digits = 3, ...)
```

Arguments

x	A CausalShrinkageForestPrediction object.
n_head	Number of observations to display per section. Default 6.
digits	Number of decimal places. Default 3.
...	Currently unused.

Value

Invisibly returns x.

See Also

[predict.CausalShrinkageForest](#), [summary.CausalShrinkageForestPrediction](#)

```
print.ShrinkageTrees
```

Print a ShrinkageTrees model

Description

Displays a concise summary of a fitted ShrinkageTrees model, including outcome type, prior, MCMC settings, acceptance ratio, and posterior mean sigma.

Usage

```
## S3 method for class 'ShrinkageTrees'
print(x, ...)
```

Arguments

x	A fitted ShrinkageTrees model object.
...	Currently unused.

Value

Invisibly returns x.

See Also

[summary.ShrinkageTrees](#), [HorseTrees](#), [ShrinkageTrees](#)

```
print.ShrinkageTreesPrediction
```

Print a ShrinkageTreesPrediction object

Description

Displays a formatted table of posterior mean predictions and credible interval bounds for the first n_head observations.

Usage

```
## S3 method for class 'ShrinkageTreesPrediction'
print(x, n_head = 6, digits = 3, ...)
```

Arguments

<code>x</code>	A <code>ShrinkageTreesPrediction</code> object.
<code>n_head</code>	Number of observations to display. Default 6.
<code>digits</code>	Number of decimal places. Default 3.
<code>...</code>	Currently unused.

Value

Invisibly returns `x`.

See Also

[predict.ShrinkageTrees](#), [summary.ShrinkageTreesPrediction](#)

```
print.summary.CausalShrinkageForest
```

Print a CausalShrinkageForest model summary

Description

Displays a detailed summary of a `CausalShrinkageForest` model, including model specification, treatment effect estimates, prognostic function, posterior sigma, variable importance for each forest, and MCMC diagnostics.

Usage

```
## S3 method for class 'summary.CausalShrinkageForest'
print(x, n_vi = 10, ...)
```

Arguments

<code>x</code>	A <code>summary.CausalShrinkageForest</code> object.
<code>n_vi</code>	Maximum number of variables to display per variable importance table. Default 10.
<code>...</code>	Currently unused.

Value

Invisibly returns `x`.

See Also

[summary.CausalShrinkageForest](#)

```
print.summary.CausalShrinkageForestPrediction
```

Print a CausalShrinkageForestPrediction summary

Description

Displays distributional summaries (min, Q1, median, max) of the posterior mean predictions and credible interval bounds, separately for the prognostic function, CATE, and total outcome.

Usage

```
## S3 method for class 'summary.CausalShrinkageForestPrediction'
print(x, digits = 3, ...)
```

Arguments

x	A summary.CausalShrinkageForestPrediction object.
digits	Number of decimal places. Default 3.
...	Currently unused.

Value

Invisibly returns x.

See Also

[summary.CausalShrinkageForestPrediction](#)

```
print.summary.ShrinkageTrees
```

Print a ShrinkageTrees model summary

Description

Displays a detailed summary of a ShrinkageTrees model, including model specification, posterior sigma, prediction summaries, variable importance, and MCMC diagnostics.

Usage

```
## S3 method for class 'summary.ShrinkageTrees'
print(x, n_vi = 10, ...)
```

Arguments

x	A summary.ShrinkageTrees object.
n_vi	Maximum number of variables to display in the variable importance table. Default 10.
...	Currently unused.

Value

Invisibly returns x.

See Also

[summary.ShrinkageTrees](#)

```
print.summary.ShrinkageTreesPrediction
```

Print a ShrinkageTreesPrediction summary

Description

Displays distributional summaries (min, Q1, median, max) of the posterior mean predictions and credible interval bounds.

Usage

```
## S3 method for class 'summary.ShrinkageTreesPrediction'
print(x, digits = 3, ...)
```

Arguments

x	A <code>summary.ShrinkageTreesPrediction</code> object.
digits	Number of decimal places. Default 3.
...	Currently unused.

Value

Invisibly returns x.

See Also

[summary.ShrinkageTreesPrediction](#)

ShrinkageTrees	<i>General Shrinkage Regression Trees (ShrinkageTrees)</i>
----------------	--

Description

Fits a Bayesian Shrinkage Tree model with flexible global-local priors on the step heights. This function generalizes [HorseTrees](#) by allowing different global-local shrinkage priors on the step heights. Supports continuous, binary, right-censored, and interval-censored outcomes.

Usage

```

ShrinkageTrees(
  y = NULL,
  status = NULL,
  X_train,
  X_test = NULL,
  left_time = NULL,
  right_time = NULL,
  outcome_type = "continuous",
  timescale = "time",
  number_of_trees = 200,
  prior_type = "horseshoe",
  local_hp = NULL,
  global_hp = NULL,
  a_dirichlet = 0.5,
  b_dirichlet = 1,
  rho_dirichlet = NULL,
  power = 2,
  base = 0.95,
  p_grow = 0.4,
  p_prune = 0.4,
  nu = 3,
  q = 0.9,
  sigma = NULL,
  N_post = 1000,
  N_burn = 1000,
  delayed_proposal = 5,
  store_posterior_sample = TRUE,
  n_chains = 1,
  verbose = TRUE
)

```

Arguments

<code>y</code>	Outcome vector. Numeric. Can represent continuous outcomes, binary outcomes (0/1), or follow-up times for survival data. Set to <code>NULL</code> when using <code>outcome_type = "interval-censored"</code> , as values are derived from <code>left_time</code> and <code>right_time</code> .
<code>status</code>	Optional censoring indicator vector (1 = event occurred, 0 = censored). Required if <code>outcome_type = "right-censored"</code> . For interval-censored outcomes, this is derived automatically from <code>left_time</code> and <code>right_time</code> .
<code>X_train</code>	Covariate matrix for training. Each row corresponds to an observation, and each column to a covariate.
<code>X_test</code>	Optional covariate matrix for test data. If <code>NULL</code> , defaults to the mean of the training covariates.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> .
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Required when <code>outcome_type = "interval-censored"</code> . Use <code>Inf</code> for right-censored observations.

outcome_type	Type of outcome. One of "continuous", "binary", "right-censored", or "interval-censored".
timescale	Indicates the scale of follow-up times. Options are "time" (nonnegative follow-up times, will be log-transformed internally) or "log" (already log-transformed). Used when outcome_type is "right-censored" or "interval-censored".
number_of_trees	Number of trees in the ensemble. Default is 200.
prior_type	Type of prior on the step heights. Options include "horseshoe", "horseshoe_fw", "half-cauchy", "standard" and "dirichlet".
local_hp	Local hyperparameter controlling shrinkage on individual step heights. Should typically be set smaller than $1 / \sqrt{\text{number_of_trees}}$. Required for prior_type = "standard".
global_hp	Global hyperparameter controlling overall shrinkage. Must be specified for Horseshoe-type priors; ignored for prior_type = "half-cauchy" or "standard".
a_dirichlet	First shape parameter of the Beta prior used in the Dirichlet-Sparse splitting rule. Together with b_dirichlet, it controls the expected sparsity level. Only when prior_type = "dirichlet".
b_dirichlet	Second shape parameter of the Beta prior for the sparsity level. Larger values shrink splitting probabilities more strongly toward uniform sparsity. Only when prior_type = "dirichlet".
rho_dirichlet	Sparsity hyperparameter. If left NULL, it defaults to the number of covariates. Only when prior_type = "dirichlet".
power	Power parameter for the tree structure prior. Default is 2.0.
base	Base parameter for the tree structure prior. Default is 0.95.
p_grow	Probability of proposing a grow move. Default is 0.4.
p_prune	Probability of proposing a prune move. Default is 0.4.
nu	Degrees of freedom for the error distribution prior. Default is 3.
q	Quantile hyperparameter for the error variance prior. Default is 0.90.
sigma	Optional known value for error standard deviation. If NULL, estimated from data.
N_post	Number of posterior samples to store. Default is 1000.
N_burn	Number of burn-in iterations. Default is 1000.
delayed_proposal	Number of delayed iterations before proposal. Only for reversible updates. Default is 5.
store_posterior_sample	Logical; whether to store posterior samples for each iteration. Default is TRUE.
n_chains	Number of independent MCMC chains to run. Default is 1 (standard single-chain behaviour). When n_chains > 1 the chains are run in parallel via <code>parallel::mclapply</code> and their posterior samples are pooled into a single ShrinkageTrees object, so all existing print, summary, and predict methods work without modification. On Windows, mclapply falls back to sequential execution.
verbose	Logical; whether to print verbose output. Default is TRUE.

Details

This function is a flexible generalization of `HorseTrees`. Instead of using a single Horseshoe prior, it allows specifying different global–local shrinkage configurations for the tree step heights. Further methodological details on the Horseshoe Forest framework can be found in Jacobs, van Wieringen & van der Pas (2025).

The horseshoe prior is the fully Bayesian global-local shrinkage prior, where both the global and local shrinkage parameters are assigned half-Cauchy distributions with scale hyperparameters `global_hp` and `local_hp`, respectively. The global shrinkage parameter is defined separately for each tree, allowing adaptive regularization per tree.

The `horseshoe_fw` prior (forest-wide horseshoe) is similar to horseshoe, except that the global shrinkage parameter is shared across all trees in the forest simultaneously.

The half-cauchy prior considers only local shrinkage and does not include a global shrinkage component. It places a half-Cauchy prior on each local shrinkage parameter with scale hyperparameter `local_hp`.

The standard prior (Chipman, George & McCulloch, 2010) corresponds to the classical BART specification, where step heights are given a normal prior with variance scaled by the number of trees. This prior does not introduce a global shrinkage parameter and does not use global–local structure.

The `dirichlet` prior implements the Dirichlet–Sparse splitting rule of Linero (2018), in which splitting probabilities follow a Dirichlet prior whose concentration is controlled by a Beta sparsity parameter (`a_dirichlet`, `b_dirichlet`) and an expected sparsity level `rho_dirichlet`.

Value

An S3 object of class "ShrinkageTrees" containing the following elements:

train_predictions Vector of posterior mean predictions on the training data.

test_predictions Vector of posterior mean predictions on the test data (or on mean covariate vector if `X_test` not provided).

sigma Vector of posterior samples of the error variance.

acceptance_ratio Average acceptance ratio across trees during sampling.

train_predictions_sample Matrix of posterior samples of training predictions (iterations in rows, observations in columns). Present only if `store_posterior_sample = TRUE`.

test_predictions_sample Matrix of posterior samples of test predictions. Present only if `store_posterior_sample = TRUE`.

train_probabilities Vector of posterior mean probabilities on the training data (only for `outcome_type = "binary"`).

test_probabilities Vector of posterior mean probabilities on the test data (only for `outcome_type = "binary"`).

train_probabilities_sample Matrix of posterior samples of training probabilities (only for `outcome_type = "binary"` and if `store_posterior_sample = TRUE`).

test_probabilities_sample Matrix of posterior samples of test probabilities (only for `outcome_type = "binary"` and if `store_posterior_sample = TRUE`).

References

Jacobs, T., van Wieringen, W. N., & van der Pas, S. L. (2025). *Horseshoe Forests for High-Dimensional Causal Survival Analysis*. arXiv:2507.22004. <https://doi.org/10.48550/arXiv.2507.22004>

Chipman, H. A., George, E. I., & McCulloch, R. E. (2010). *BART: Bayesian additive regression trees*. *Annals of Applied Statistics*.

Linero, A. R. (2018). *Bayesian regression trees for high-dimensional prediction and variable selection*. *Journal of the American Statistical Association*.

See Also

Model family: [HorseTrees](#) (horseshoe prior), [CausalHorseForest](#) (causal inference), [CausalShrinkageForest](#) (causal, flexible prior).

Survival wrappers: [SurvivalBART](#), [SurvivalDART](#).

S3 methods: [print.ShrinkageTrees](#), [summary.ShrinkageTrees](#), [predict.ShrinkageTrees](#), [plot.ShrinkageTrees](#).

Examples

```
# Example: Continuous outcome with ShrinkageTrees, two priors
n <- 50
p <- 3
X <- matrix(runif(n * p), ncol = p)
X_test <- matrix(runif(n * p), ncol = p)
y <- X[, 1] + rnorm(n)

# Fit ShrinkageTrees with standard horseshoe prior
fit_horseshoe <- ShrinkageTrees(y = y,
                               X_train = X,
                               X_test = X_test,
                               outcome_type = "continuous",
                               number_of_trees = 5,
                               prior_type = "horseshoe",
                               local_hp = 0.1 / sqrt(5),
                               global_hp = 0.1 / sqrt(5),
                               N_post = 10,
                               N_burn = 5,
                               store_posterior_sample = TRUE,
                               verbose = FALSE)

# Fit ShrinkageTrees with half-Cauchy prior
fit_halfcauchy <- ShrinkageTrees(y = y,
                                 X_train = X,
                                 X_test = X_test,
                                 outcome_type = "continuous",
                                 number_of_trees = 5,
                                 prior_type = "half-cauchy",
                                 local_hp = 1 / sqrt(5),
                                 N_post = 10,
                                 N_burn = 5,
                                 store_posterior_sample = TRUE,
                                 verbose = FALSE)

# Posterior mean predictions
pred_horseshoe <- colMeans(fit_horseshoe$train_predictions_sample)
pred_halfcauchy <- colMeans(fit_halfcauchy$train_predictions_sample)

# Posteriors of the mean (global average prediction)
post_mean_horseshoe <- rowMeans(fit_horseshoe$train_predictions_sample)
```



```

post_mean_halfcauchy <- rowMeans(fit_halfcauchy$train_predictions_sample)

# Posterior mean prediction averages
mean_pred_horseshoe <- mean(post_mean_horseshoe)
mean_pred_halfcauchy <- mean(post_mean_halfcauchy)

# Example: Interval-censored survival outcome
n <- 50; p <- 3
X_ic <- matrix(rnorm(n * p), ncol = p)
true_t <- rexp(n, rate = exp(-X_ic[, 1]))
left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
exact <- sample(n, 15)
left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 10); right_t[rc] <- Inf

fit_ic <- ShrinkageTrees(left_time = left_t, right_time = right_t,
                        X_train = X_ic,
                        outcome_type = "interval-censored",
                        prior_type = "horseshoe",
                        local_hp = 0.1 / sqrt(5),
                        global_hp = 0.1 / sqrt(5),
                        number_of_trees = 5,
                        N_post = 10, N_burn = 5,
                        verbose = FALSE)

```

summary.CausalShrinkageForest

Summarise a CausalShrinkageForest model

Description

Returns an inspectable list with treatment effect estimates, prognostic function summaries, posterior sigma, variable importance for each forest, and MCMC diagnostics.

Usage

```

## S3 method for class 'CausalShrinkageForest'
summary(object, bayesian_bootstrap = TRUE, ...)

```

Arguments

object	A fitted CausalShrinkageForest model object.
bayesian_bootstrap	Logical; if TRUE (default), the ATE posterior is computed by reweighting the per-iteration CATE vector with Dirichlet(1, ..., 1) weights (Bayesian bootstrap). This gives a draw from the posterior of the <i>population</i> ATE (PATE), with a credible interval that accounts for uncertainty in the covariate distribution. If FALSE, equal $1/n$ weights are used, giving the mixed ATE (MATE), which conditions on the observed covariates. Ignored when posterior samples are not stored.
...	Currently unused.

Value

A summary.CausalShrinkageForest object with elements:

call The original model call.

outcome_type Outcome type.

timescale Timescale for survival outcomes.

prior Prior specification for control and treatment forests.

mcmc MCMC settings.

data_info Training and test data dimensions.

treatment_effect List with ate (posterior mean ATE), cate_sd (SD of individual CATEs), and optionally ate_lower, ate_upper (95 percent credible interval; requires store_posterior_sample = TRUE) and bayesian_bootstrap (the flag used to produce the CI).

prognostic Summary of the prognostic function (mean, SD, range).

sigma Named vector with posterior mean, SD, and 95 percent credible interval of sigma (if estimated).

variable_importance_control Variable importance for the control forest (if available).

variable_importance_treat Variable importance for the treatment forest (if available).

acceptance_ratios List with acceptance ratios for each forest.

See Also

[print.summary.CausalShrinkageForest](#), [CausalHorseForest](#), [CausalShrinkageForest](#)

summary.CausalShrinkageForestPrediction

Summarise a CausalShrinkageForestPrediction object

Description

Returns distributional summaries (min, Q1, median, max) of the posterior mean predictions and credible interval bounds across all test observations, separately for the prognostic function, CATE, and total outcome.

Usage

```
## S3 method for class 'CausalShrinkageForestPrediction'
summary(object, ...)
```

Arguments

object	A CausalShrinkageForestPrediction object.
...	Currently unused.

Value

A summary.CausalShrinkageForestPrediction object.

See Also

[predict.CausalShrinkageForest](#), [print.summary.CausalShrinkageForestPrediction](#)

summary.ShrinkageTrees

Summarise a ShrinkageTrees model

Description

Returns an inspectable list with posterior sigma summaries, prediction summaries, variable importance (posterior inclusion probabilities), and MCMC diagnostics.

Usage

```
## S3 method for class 'ShrinkageTrees'
summary(object, ...)
```

Arguments

<code>object</code>	A fitted ShrinkageTrees model object.
<code>...</code>	Currently unused.

Value

A summary.ShrinkageTrees object with elements:

call The original model call.

outcome_type Outcome type ("continuous", "binary", "right-censored", or "interval-censored").

timescale Timescale for survival outcomes ("time" or "log").

prior Prior specification.

mcmc MCMC settings (trees, draws, burn-in).

data_info Training and test data dimensions.

sigma Named vector with posterior mean, SD, and 95 percent credible interval of sigma (continuous and survival outcomes only).

predictions List with train (and optionally test) prediction summaries (mean, SD, range).

variable_importance Named vector of posterior inclusion probabilities, sorted decreasingly (if available).

acceptance_ratio MCMC acceptance ratio vector.

diagnostics (When **coda** is installed) A list with ess (effective sample size) and, for multi-chain fits, rhat (Gelman–Rubin $\hat{R}$).

See Also

[print.summary.ShrinkageTrees](#), [as.mcmc.list.ShrinkageTrees](#), [HorseTrees](#), [ShrinkageTrees](#)

summary.ShrinkageTreesPrediction

Summarise a ShrinkageTreesPrediction object

Description

Returns distributional summaries (min, Q1, median, max) of the posterior mean predictions and credible interval bounds across all observations.

Usage

```
## S3 method for class 'ShrinkageTreesPrediction'
summary(object, ...)
```

Arguments

object	A ShrinkageTreesPrediction object.
...	Currently unused.

Value

A summary.ShrinkageTreesPrediction object.

See Also

[predict.ShrinkageTrees](#), [print.summary.ShrinkageTreesPrediction](#)

SurvivalBART

SurvivalBART

Description

Fits an Accelerated Failure Time (AFT) model using the classical Bayesian Additive Regression Trees (BART) prior: $\log(Y) = f(x) + \varepsilon$. Supports both right-censored and interval-censored survival outcomes.

Usage

```
SurvivalBART(
  time = NULL,
  status = NULL,
  X_train,
  X_test = NULL,
  timescale = "time",
  number_of_trees = 200,
  k = 2,
  N_post = 1000,
  N_burn = 1000,
  store_posterior_sample = TRUE,
  verbose = TRUE,
```

```

    left_time = NULL,
    right_time = NULL,
    ...
)

```

Arguments

<code>time</code>	Outcome vector of (non-negative) survival times. Required for right-censored outcomes; set to NULL when using interval censoring.
<code>status</code>	Event indicator (1 = event, 0 = censored). Required for right-censored outcomes; derived automatically for interval censoring.
<code>X_train</code>	Design matrix for training data.
<code>X_test</code>	Optional test matrix. If NULL, predictions are computed at the column means of <code>X_train</code> .
<code>timescale</code>	Either "time" (log-transform internally) or "log" (already log-transformed).
<code>number_of_trees</code>	Number of trees in the ensemble. Default is 200.
<code>k</code>	Scaling constant used to calibrate the prior variance of the step heights.
<code>N_post</code>	Number of posterior samples to store.
<code>N_burn</code>	Number of burn-in iterations.
<code>store_posterior_sample</code>	Logical; if TRUE (default), store the full $N_{\text{post}} \times n$ matrix of posterior predictions. Required for <code>predict()</code> , <code>plot(type = "survival")</code> with full posterior credible bands, and custom posterior analyses. Set to FALSE to save memory when only posterior means are needed.
<code>verbose</code>	Logical; print sampling progress.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries for interval-censored data. Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> . When provided together with <code>right_time</code> , the model is fitted with <code>outcome_type = "interval-censored"</code> and <code>time/status</code> are ignored.
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Use <code>Inf</code> for right-censored observations.
<code>...</code>	Additional arguments passed to ShrinkageTrees to override default hyperparameters.

Details

This function provides a survival-specific interface for classical BART under an AFT formulation for right-censored or interval-censored outcomes.

For right-censored data, supply `time` and `status`. For interval-censored data, supply `left_time` and `right_time` instead; event indicators are derived internally following the `survival::Surv(type = "interval2")` convention.

Structural regularisation is induced through the standard Gaussian leaf prior and tree depth prior of Chipman, George & McCulloch (2010).

Users requiring alternative shrinkage priors (e.g., Horseshoe or Dirichlet splitting priors) should use [ShrinkageTrees](#) directly.

Value

An object of class "ShrinkageTrees" fitted under a classical BART prior within an AFT formulation.

See [ShrinkageTrees](#) for a full description of returned components

References

Chipman, H. A., George, E. I., & McCulloch, R. E. (2010). Bayesian Additive Regression Trees. *Annals of Applied Statistics*.

See Also

Related models: [SurvivalDART](#) (Dirichlet sparsity), [HorseTrees](#) (horseshoe prior), [ShrinkageTrees](#) (general shrinkage priors).

S3 methods: [print.ShrinkageTrees](#), [summary.ShrinkageTrees](#), [predict.ShrinkageTrees](#), [plot.ShrinkageTrees](#).

Examples

```
set.seed(1)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
time <- rexp(n, rate = exp(0.5 * X[, 1]))
status <- rbinom(n, 1, 0.7)

fit <- SurvivalBART(time = time, status = status, X_train = X,
                    number_of_trees = 5, N_post = 50, N_burn = 25,
                    verbose = FALSE)

# S3 methods
print(fit)
smry <- summary(fit)

# Posterior predictions on new data
X_new <- matrix(rnorm(10 * p), ncol = p)
pred <- predict(fit, newdata = X_new)
print(pred)

# Diagnostic plot (requires ggplot2)
if (requireNamespace("ggplot2", quietly = TRUE)) {
  plot(fit, type = "trace")

  # Posterior survival curves for training data
  plot(fit, type = "survival")

  # Posterior predictive survival curves for new data
  plot(pred, type = "survival")
  plot(pred, type = "survival", obs = c(1, 5))
}

# Interval-censored example
set.seed(11)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
true_t <- rexp(n, rate = exp(0.5 * X[, 1]))
```

```

left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
exact <- sample(n, 10); left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 5); right_t[rc] <- Inf

fit_ic <- SurvivalBART(left_time = left_t, right_time = right_t,
                      X_train = X, number_of_trees = 5,
                      N_post = 50, N_burn = 25, verbose = FALSE)

```

SurvivalBCF

*SurvivalBCF (Bayesian Causal Forest for survival data)***Description**

Fits an Accelerated Failure Time (AFT) version of Bayesian Causal Forest (BCF): $Y = \mu(x) + W\tau(x) + \varepsilon$, where separate forests are used for the prognostic (control) function $\mu(x)$ and the treatment effect function $\tau(x)$.

Usage

```

SurvivalBCF(
  time = NULL,
  status = NULL,
  X_train,
  treatment,
  timescale = "time",
  propensity = NULL,
  treatment_coding = "centered",
  number_of_trees_control = 200,
  number_of_trees_treat = 50,
  power_control = 2,
  base_control = 0.95,
  power_treat = 3,
  base_treat = 0.25,
  N_post = 1000,
  N_burn = 1000,
  store_posterior_sample = TRUE,
  verbose = TRUE,
  left_time = NULL,
  right_time = NULL,
  ...
)

```

Arguments

time	Outcome vector of (non-negative) survival times. Required for right-censored outcomes; set to NULL when using interval censoring.
status	Event indicator (1 = event, 0 = censored). Required for right-censored outcomes; derived automatically for interval censoring.
X_train	Design matrix for training data.

treatment	Treatment indicator (0/1) for training data.
timescale	Either "time" (log-transform internally) or "log" (already log-transformed).
propensity	Optional vector of propensity scores. If provided, it is appended to the control forest design matrix. Required when treatment_coding = "adaptive".
treatment_coding	Character string specifying how the treatment indicator enters the model. One of "centered" (default, maps to $-1/2$ and $1/2$), "binary" (maps to 0 and 1), "adaptive" (maps to $z_i - \hat{e}(x_i)$, where $\hat{e}(x_i)$ is the propensity score), or "invariant" (parameter-expanded coding with $b_0, b_1 \sim N(0, 1/2)$ estimated within the Gibbs sampler; Hahn et al., 2020, Section 5.2).
number_of_trees_control	Number of trees in the control forest. Default is 200.
number_of_trees_treat	Number of trees in the treatment forest. Default is 50.
power_control, base_control	Tree-structure prior parameters for the control forest.
power_treat, base_treat	Tree-structure prior parameters for the treatment forest.
N_post	Number of posterior samples to store.
N_burn	Number of burn-in iterations.
store_posterior_sample	Logical; if TRUE (default), store the full $N_{\text{post}} \times n$ matrix of posterior predictions. Required for predict(), plot(type = "survival") with full posterior credible bands, and custom posterior analyses. Set to FALSE to save memory when only posterior means are needed.
verbose	Logical; print sampling progress.
left_time	Optional numeric vector of left (lower) time boundaries for interval-censored data. Exact events have left_time == right_time; right-censored observations have right_time = Inf; interval-censored observations have finite left_time < right_time. When provided together with right_time, the model is fitted with outcome_type = "interval-censored" and time/status are ignored.
right_time	Optional numeric vector of right (upper) time boundaries. Use Inf for right-censored observations.
...	Additional arguments passed to CausalShrinkageForest to override default hyperparameters.

Details

This wrapper provides a survival-specific implementation using classical BART-style priors for both forests. Supports both right-censored and interval-censored survival outcomes.

This function implements a simplified AFT-BCF model for right-censored or interval-censored survival outcomes. Structural regularisation is induced through classical BART priors on the tree structure and leaf parameters.

For right-censored data, supply time and status. For interval-censored data, supply left_time and right_time instead; event indicators are derived internally following the `survival::Surv(type = "interval2")` convention.

Users requiring alternative shrinkage priors (e.g., Horseshoe or Dirichlet splitting priors) should use [SurvivalShrinkageBCF](#) or call [CausalShrinkageForest](#) directly.

Value

An object of class "CausalShrinkageForest" corresponding to a survival BCF model under classical BART priors.

See [CausalShrinkageForest](#) for returned components.

References

Hahn, P. R., Murray, J. S., & Carvalho, C. M. (2020). Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects. *Bayesian Analysis*.

See Also

Related models: [SurvivalShrinkageBCF](#) (Dirichlet sparsity), [CausalHorseForest](#) (horseshoe prior), [CausalShrinkageForest](#) (general shrinkage priors).

S3 methods: [print.CausalShrinkageForest](#), [summary.CausalShrinkageForest](#), [predict.CausalShrinkageForest](#), [plot.CausalShrinkageForest](#).

Examples

```
set.seed(3)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
treatment <- rbinom(n, 1, 0.5)
log_T <- X[, 1] + treatment * (-0.5) + rnorm(n)
time <- exp(log_T)
status <- rbinom(n, 1, 0.7)

fit <- SurvivalBCF(time = time, status = status, X_train = X,
                  treatment = treatment,
                  number_of_trees_control = 5,
                  number_of_trees_treat = 5,
                  N_post = 50, N_burn = 25,
                  verbose = FALSE)

# S3 methods
print(fit)
smry <- summary(fit)

# Posterior ATE
cat("ATE:", round(smry$treatment_effect$ate, 3), "\n")

# Diagnostic and treatment-effect plots (requires ggplot2)
if (requireNamespace("ggplot2", quietly = TRUE)) {
  plot(fit, type = "trace")
  plot(fit, type = "ate")
  plot(fit, type = "cate")
}

# Interval-censored causal example
set.seed(13)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
treatment <- rbinom(n, 1, 0.5)
true_t <- exp(X[, 1] + treatment * (-0.5) + rnorm(n))
left_t <- true_t * runif(n, 0.5, 1)
```

```

right_t <- true_t * runif(n, 1, 1.5)
exact <- sample(n, 10); left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 5); right_t[rc] <- Inf

fit_ic <- SurvivalBCF(left_time = left_t, right_time = right_t,
                     X_train = X, treatment = treatment,
                     number_of_trees_control = 5,
                     number_of_trees_treat = 5,
                     N_post = 50, N_burn = 25, verbose = FALSE)

```

SurvivalDART

SurvivalDART

Description

Fits an Accelerated Failure Time (AFT) model using the Dirichlet splitting prior (DART), which induces structural sparsity through a Beta-Dirichlet hierarchy on splitting probabilities. Supports both right-censored and interval-censored survival outcomes.

Usage

```

SurvivalDART(
  time = NULL,
  status = NULL,
  X_train,
  X_test = NULL,
  timescale = "time",
  number_of_trees = 200,
  a_dirichlet = 0.5,
  b_dirichlet = 1,
  rho_dirichlet = NULL,
  k = 2,
  N_post = 1000,
  N_burn = 1000,
  store_posterior_sample = TRUE,
  verbose = TRUE,
  left_time = NULL,
  right_time = NULL,
  ...
)

```

Arguments

time	Outcome vector of (non-negative) survival times. Required for right-censored outcomes; set to NULL when using interval censoring.
status	Event indicator (1 = event, 0 = censored). Required for right-censored outcomes; derived automatically for interval censoring.
X_train	Design matrix for training data.
X_test	Optional test matrix. If NULL, predictions are computed at the column means of X_train.

timescale	Either "time" (log-transform internally) or "log" (already log-transformed).
number_of_trees	Number of trees in the ensemble. Default is 200.
a_dirichlet, b_dirichlet	Beta hyperparameters controlling sparsity in the Dirichlet splitting rule.
rho_dirichlet	Expected number of active predictors. If NULL, defaults to the number of co-variates in X_{train} .
k	Scaling constant used to calibrate the prior variance of the step heights.
N_post	Number of posterior samples to store.
N_burn	Number of burn-in iterations.
store_posterior_sample	Logical; if TRUE (default), store the full $N_{\text{post}} \times n$ matrix of posterior predictions. Required for <code>predict()</code> , <code>plot(type = "survival")</code> with full posterior credible bands, and custom posterior analyses. Set to FALSE to save memory when only posterior means are needed.
verbose	Logical; print sampling progress.
left_time	Optional numeric vector of left (lower) time boundaries for interval-censored data. Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> . When provided together with <code>right_time</code> , the model is fitted with <code>outcome_type = "interval-censored"</code> and <code>time/status</code> are ignored.
right_time	Optional numeric vector of right (upper) time boundaries. Use <code>Inf</code> for right-censored observations.
...	Additional arguments passed to ShrinkageTrees to override default hyperparameters.

Details

This function provides a survival-specific wrapper for DART under an AFT formulation for right-censored or interval-censored outcomes.

For right-censored data, supply `time` and `status`. For interval-censored data, supply `left_time` and `right_time` instead; event indicators are derived internally following the `survival::Surv(type = "interval2")` convention.

Structural regularisation is induced through a Dirichlet prior on splitting probabilities, encouraging sparse feature usage in high-dimensional settings.

Users requiring alternative shrinkage priors on the leaf parameters (e.g., Horseshoe or half-Cauchy priors) should use [ShrinkageTrees](#) directly.

Value

An object of class "ShrinkageTrees" fitted under a Dirichlet splitting prior (DART) within an AFT formulation.

See [ShrinkageTrees](#) for a full description of returned components.

See Also

Related models: [SurvivalBART](#) (standard BART prior), [ShrinkageTrees](#) (general shrinkage priors).

S3 methods: [print.ShrinkageTrees](#), [summary.ShrinkageTrees](#), [predict.ShrinkageTrees](#), [plot.ShrinkageTrees](#).

Examples

```

set.seed(2)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
time <- rexp(n, rate = exp(0.5 * X[, 1]))
status <- rbinom(n, 1, 0.7)

fit <- SurvivalDART(time = time, status = status, X_train = X,
                   number_of_trees = 5, N_post = 50, N_burn = 25,
                   verbose = FALSE)

# S3 methods
print(fit)
smry <- summary(fit)

# Posterior predictions on new data
X_new <- matrix(rnorm(10 * p), ncol = p)
pred <- predict(fit, newdata = X_new)
print(pred)

# Variable importance and survival plots (requires ggplot2)
if (requireNamespace("ggplot2", quietly = TRUE)) {
  plot(fit, type = "vi", n_vi = 5)

  # Posterior survival curves for training data
  plot(fit, type = "survival")

  # Posterior predictive survival curves for new data
  plot(pred, type = "survival")
  plot(pred, type = "survival", obs = c(1, 5))
}

# Interval-censored example
set.seed(12)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
true_t <- rexp(n, rate = exp(0.5 * X[, 1]))
left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
exact <- sample(n, 10); left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 5); right_t[rc] <- Inf

fit_ic <- SurvivalDART(left_time = left_t, right_time = right_t,
                      X_train = X, number_of_trees = 5,
                      N_post = 50, N_burn = 25, verbose = FALSE)

```

SurvivalShrinkageBCF	<i>SurvivalShrinkageBCF (Shrinkage Bayesian Causal Forest for survival data)</i>
----------------------	--

Description

Fits a survival version of a Bayesian Causal Forest (BCF) under an accelerated failure time (AFT) model, combining Dirichlet splitting priors with global-local shrinkage. Supports both right-censored

and interval-censored survival outcomes.

Usage

```
SurvivalShrinkageBCF(
  time = NULL,
  status = NULL,
  X_train,
  treatment,
  timescale = "time",
  propensity = NULL,
  treatment_coding = "centered",
  a_dir = 0.5,
  b_dir = 1,
  number_of_trees_control = 200,
  number_of_trees_treat = 50,
  power_control = 2,
  base_control = 0.95,
  power_treat = 3,
  base_treat = 0.25,
  N_post = 1000,
  N_burn = 1000,
  store_posterior_sample = TRUE,
  verbose = TRUE,
  left_time = NULL,
  right_time = NULL,
  ...
)
```

Arguments

time	Outcome vector of (non-negative) survival times. Required for right-censored outcomes; set to NULL when using interval censoring.
status	Event indicator (1 = event, 0 = censored). Required for right-censored outcomes; derived automatically for interval censoring.
X_train	Design matrix for training data.
treatment	Treatment indicator (0/1) for training data.
timescale	Either "time" (log-transform internally) or "log" (already log-transformed).
propensity	Optional vector of propensity scores. If provided, it is appended to the control forest design matrix. Required when treatment_coding = "adaptive".
treatment_coding	Character string specifying how the treatment indicator enters the model. One of "centered" (default, maps to $-1/2$ and $1/2$), "binary" (maps to 0 and 1), "adaptive" (maps to $z_i - \hat{e}(x_i)$, where $\hat{e}(x_i)$ is the propensity score), or "invariant" (parameter-expanded coding with $b_0, b_1 \sim N(0, 1/2)$ estimated within the Gibbs sampler; Hahn et al., 2020, Section 5.2).
a_dir	First shape parameter of the Beta prior controlling the sparsity level in the Dirichlet splitting rule.
b_dir	Second shape parameter of the Beta prior controlling the sparsity level in the Dirichlet splitting rule.

<code>number_of_trees_control</code>	Number of trees in the control forest. Default is 200.
<code>number_of_trees_treat</code>	Number of trees in the treatment forest. Default is 50.
<code>power_control, base_control</code>	Tree-structure prior parameters for the control forest.
<code>power_treat, base_treat</code>	Tree-structure prior parameters for the treatment forest.
<code>N_post</code>	Number of posterior samples to store.
<code>N_burn</code>	Number of burn-in iterations.
<code>store_posterior_sample</code>	Logical; if TRUE (default), store the full $N_{\text{post}} \times n$ matrix of posterior predictions. Required for <code>predict()</code> , <code>plot(type = "survival")</code> with full posterior credible bands, and custom posterior analyses. Set to FALSE to save memory when only posterior means are needed.
<code>verbose</code>	Logical; print sampling progress.
<code>left_time</code>	Optional numeric vector of left (lower) time boundaries for interval-censored data. Exact events have <code>left_time == right_time</code> ; right-censored observations have <code>right_time = Inf</code> ; interval-censored observations have finite <code>left_time < right_time</code> . When provided together with <code>right_time</code> , the model is fitted with <code>outcome_type = "interval-censored"</code> and <code>time/status</code> are ignored.
<code>right_time</code>	Optional numeric vector of right (upper) time boundaries. Use <code>Inf</code> for right-censored observations.
<code>...</code>	Additional arguments passed to CausalShrinkageForest to override default hyperparameters.

Details

This wrapper extends [SurvivalBCF](#) by incorporating Dirichlet sparsity in both the prognostic (control) and treatment forests, while applying additional shrinkage to the control forest via a half-Cauchy prior.

The SurvivalShrinkageBCF model decomposes the outcome as

$$\log T = \mu(x) + a \cdot \tau(x) + \varepsilon,$$

where $\mu(x)$ represents the prognostic (control) component and $\tau(x)$ the heterogeneous treatment effect.

In contrast to `SurvivalBCF`, this function:

- Applies a Dirichlet splitting prior to both forests, inducing structural sparsity in variable selection.
- Combines Dirichlet sparsity with additional half-Cauchy shrinkage in the control forest.

The Dirichlet prior follows the sparse splitting framework of Linero (2018), where splitting probabilities are governed by a Beta-Dirichlet hierarchy. The sparsity level is controlled by `a_dir` and `b_dir`.

Survival outcomes are modeled using an AFT formulation with censoring handled via data augmentation. Both right-censored and interval-censored data are supported. For interval-censored data, supply `left_time` and `right_time` instead of `time` and `status`.

Value

An object of class "CausalShrinkageForest" fitted with Dirichlet splitting priors and additional shrinkage.

References

Caron, A., Baio, G., & Manolopoulou, I. (2022). Shrinkage Bayesian Causal Forests for Heterogeneous Treatment Effects Estimation. *Journal of Computational and Graphical Statistics*, 31(4), 1202–1214. <https://doi.org/10.1080/10618600.2022.2067549>

See Also

Related models: [SurvivalBCF](#) (standard BART priors), [CausalShrinkageForest](#) (general shrinkage priors), [CausalHorseForest](#) (horseshoe prior).

S3 methods: [print.CausalShrinkageForest](#), [summary.CausalShrinkageForest](#), [predict.CausalShrinkageForest](#), [plot.CausalShrinkageForest](#).

Examples

```
set.seed(4)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
treatment <- rbinom(n, 1, 0.5)
log_T <- X[, 1] + treatment * (-0.5) + rnorm(n)
time <- exp(log_T)
status <- rbinom(n, 1, 0.7)

fit <- SurvivalShrinkageBCF(time = time, status = status, X_train = X,
                           treatment = treatment,
                           number_of_trees_control = 5,
                           number_of_trees_treat = 5,
                           N_post = 50, N_burn = 25,
                           verbose = FALSE)

# S3 methods
print(fit)
smry <- summary(fit)

# Posterior ATE with 95% credible interval
cat("ATE:", round(smry$treatment_effect$ate, 3), "\n")

# Diagnostic and treatment-effect plots (requires ggplot2)
if (requireNamespace("ggplot2", quietly = TRUE)) {
  plot(fit, type = "trace")
  plot(fit, type = "cate")
}

# Interval-censored causal example
set.seed(14)
n <- 30; p <- 5
X <- matrix(rnorm(n * p), ncol = p)
treatment <- rbinom(n, 1, 0.5)
true_t <- exp(X[, 1] + treatment * (-0.5) + rnorm(n))
left_t <- true_t * runif(n, 0.5, 1)
right_t <- true_t * runif(n, 1, 1.5)
```

```
exact <- sample(n, 10); left_t[exact] <- true_t[exact]; right_t[exact] <- true_t[exact]
rc <- sample(setdiff(seq_len(n), exact), 5); right_t[rc] <- Inf

fit_ic <- SurvivalShrinkageBCF(left_time = left_t, right_time = right_t,
                              X_train = X, treatment = treatment,
                              number_of_trees_control = 5,
                              number_of_trees_treat = 5,
                              N_post = 50, N_burn = 25, verbose = FALSE)
```